Daily Drilling Report
SPD19B-03
9/11/2014

PETROPARS Oil & Gas Developer

N.I.O.C. Pars Oil & Gas Company

Well Data

Well Name	Operator	Field Name	Well Type 1	Rig	Code
SPD19B-03	Petropars Ltd.	South Pars Gas Field	Development	Deep Driller, DD6	SP19-3B-01-DL-FOR-0...
Report Number	**Hole Section**	**Days LTI (days)**	**Water Depth (m)**	**Original KB Elevation (m)**	**Head Count**
46.0	8 1/2"	191.00	72.83	31.70	100.0
Last Cas	**Daily Cost Total**	**Daily Mud Cost**	**Cum Cost To Date**	**AFE Number**	**Cum Time Log Days (days)**
9 5/8" Shoe, 2,720.80mKB	264,057.41		11,313,130.42		41.90
Depth Start (mKB)	**Depth End (mKB)**	**Depth End (TVD) (mKB)**	**Depth Progress (m)**	**Drilling Hours (hrs)**	**Average ROP (m/hr)**
3,232.00	3,232.00	0.00			

Operations at Report Time
Performing full set wireline logging job.

Operations Summary
B/Ream f/ 2950m to 2717m. RIH back to bottom at 3232m section TD. Sweep the hole clean. Pump out of the hole 400m inside 9 5/8" casing to 2370m. POOH on elevator to surface. R/U wireline logging sheaves. RIH logging tool string (Resistivity + Density + Neutron + Orientation + SGR) on wireline to 400m.

Operations next 24hrs
Perform wireline logging. RIH 7" Liner.

Supervisors Contact

Job Contact	Position
Mojtaba Parvaresh	Drilling Supt.
Mohammadreza Asaie	Senior Drilling Sup.
Behnam Jahangiri	Night Drilling Sup.
Behzad Khalilmoghadam	Drilling Engineer
Ali Sahebnouri	Geologist
R. Kamalpour	POGC Rep.

HSE Data

Type	Last Date	Next Date
Abandon Drill	8/23/2014	10/13/2014
BOP Test	9/4/2014	9/26/2014
Chemical Spill Drill	8/17/2014	10/17/2014
Fire Drill	8/23/2014	10/13/2014
H2S Drill	9/6/2014	10/7/2014
Heli Crash Drill	7/15/2014	9/14/2014
Medical Drill	7/26/2014	9/25/2014
MOB Drill	9/1/2014	11/1/2014
Pit Drill	9/10/2014	10/26/2014
Trip Drill	9/11/2014	11/1/2014

STOP Cards

Number of Reports: 34

Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
00:00	02:30	2.50	8 1/2"	D	06	P	B/Ream f/ 2950m to 2720m (9 5/8'' CSG Shoe) w/ 450 gpm / 1550 psi / 40 rpm / 13-17 Kft-lbs.
02:30	05:00	2.50	8 1/2"	D	04	P	Flow check inside shoe, well static. RIH on elevator f/ 2720m to bottom (3232m) in good hole condition.
05:00	06:30	1.50	8 1/2"	D	07	P	Sweep the hole by pumping 20 bbl Lo-Vis + 20 bbl Hi-Vis tandem pill. Circulate out w/ 550 gpm / 2250 psi / 15 rpm. Meanwhile keep the string in motion. Confirm the shakers are clean.
06:30	14:30	8.00	8 1/2"	D	09	P	Pump 140 bbl Hi-Vis pill and displace it to reach the bit w/ 550 gpm/ 2200 psi. Start pumping out of the hole with 50 gpm f/ 3232m to 2720m 9 5/8" CSG shoe (Displace 7 bbl Hi-Vis pill out of the bit per stand). Cont' Pumping out of the hole with 350 gpm / 950 psi f/ 2720m to 2340m.
14:30	21:30	7.00	8 1/2"	D	04	P	Normal POOH drilling BHA#11 f/ 2340m to surface. L/D 7" Jar, R/B HWDPs, 6 3/4" NMDC, Float sub, X/O sub, B/O & L/D Tx sub (Retrieve MWD probe). Drain the motor, B/O the bit & R/B motor, 8" stab. & NMDC in derrick. ** 8 1/2" PDC bit dull grading: 1-1-WT-G/S-X-1/8-NO-TD ** Motor bearing gap: 3 mm ** Sleeve stabilizer & String Stabilizer: in gauge
21:30	22:00	0.50	8 1/2"	J	01	P	HPJSM. R/U PDK wireline logging sheaves.
22:00	23:30	1.50	8 1/2"	J	04	P	P/U and M/U wireline logging tool string as per below, RIH to 24.6m. (Total Weight: 689.3 kg, Tools Max OD: 4.76") ITEMS: — Length (m) Bull Plug — 0.045 MSFL — 3.46 DLL (with 7" Stand off Centralizer) — 6.24 Iso Sub — 0.643 Knuckle Joint — 0.771 PE Density — 3.284 Compensated Neutron — 2.301 Spectrum Gamma — 2.026 Orientation — 2.127 Telemetry (with 7" Stand off Centralizer) — 1.55 Iso Sub — 0.643 Swivel — 1.515

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Operations Summary

Start Time	End Time	Dur (hrs)	Phase	Code 1	Code 2	Time P-N-T	Comment
23:30	00:00	0.50	8 1/2"	J	04	P	RIH logging BHA on wireline to 400m.
00:00	01:00	1.00	8 1/2"				Sep 12, 2014 00:00-01:00 RIH logging BHA on wireline f/ 400m to 3232m 8 1/2" section TD ** Repeat Log f/ 2850m to 2721m. ** While RIH at 3000m string weight dropped to 1500 lbs (Normal D/Weight 2300 lbs) . Pulled the string to 5500 lbs (Normal U/weight 4800 lbs). keep the tension for 10 min, no success. Increase over pull to 6000 lbs, string got free. 03:00-04:30 Start logging up f/ 3232 to 2890m. While logging up obs line tension increase f/ 4800 lbs to 5200 lbs and tool string stuck at 2890m. 04:30-06:00 Work on tool string, in progress.

Mud Information

Depth (mKB)	Type	Density (lb/gal)	Funnel Viscosity (s/qt)	PV Calc (cp)	YP Calc (lbf/100ft²)	Filtrate (mL/30min)
3,232.00	CaCo3/KCL/ Polymer	10.50	55	24.0	21.000	2.8
pH	**Pf (mL/mL)**	**Mf (mL/mL)**	**Pm (mL/mL)**	**Sand (%)**	**Low Gravity Solids (%)**	**High Gravity Solids (%)**
10.5	0.500	1.400	1.200			
Vis 3rpm	**Vis 6rpm**	**Percent Oil (%)**	**Percent Water (%)**	**Solids (%)**	**Chlorides (mg/L)**	**Filter Cake (/32")**
5	7	0.0	87.0	13.0	33,000.000	1
Gel 10 sec (lbf/100ft²)	**Gel 10 min (lbf/100ft²)**	**Gel 30 min (lbf/100ft²)**	**Oil Water Ratio**	**T (fl) (°C)**	**Active Mud Volume (bbl)**	**Reserve Mud Volume (bbl)**
6.000	8.000	11.000	0/100		873.0	2415.0
Cum Mud Added (bbl)	**Mud Lost to Surface (bbl)**	**Mud Lost to Hole (bbl)**	**Avg Background Gas (%)**	**Avg Drill Gas (%)**	**K**	**n**
7589.0	50.0	0.0			13.39637	0.396

Marine Conditions

Swell Height (m)	Visibility (km)	Wind Direction (°)	Wind Speed (knots)	Barometric Pressure (mmHg)	Temperature - High (°C)
1.50	5.	340	15.0	1,000.0	36.0

Support Vessels

Vessel Name	Type	Arrival Date	Depart	Note
EP-PGD	Chopper			Pax in: 0 Pax out: 5
EP-HBJ	Chopper			Pax in: 7 Pax out: 5
Marina 1	Supply Boat			Streaming at Loc.
Ahmed 6	Supply Boat	9/7/2014 17:30		C/Out Personnel Safety watch & Chopper Safety watch, Received 20 MT FO f/ BORKAN at SR-2 anchorag
Moje Darya	Supply Boat			Streaming at Loc.
Borkan	Supply Boat	9/11/2014 19:15	9/11/2014 21:00	Arrived f/ SR2 & A/S DD6, B/L cargo for Aban8. Transferred FO-10Mt to Marina-1. Dep for Aban 8 at 21

Mud Additives

Description	Consumed	Cum Rec	On Loc

Bulk Amounts

Supply Item Description	Consumed	Received	Cum On Loc
Potable Water	22.0	18.0	164.0
Cement G	0.0	0.0	92.0
Drill Water	10.0	0.0	294.0
Diesel Fuel	11.0	0.0	134.0
Barite Bulk	0.0	0.0	129.0
Barite BB	0.0	0.0	0.0
Bentonite Bulk	0.0	0.0	8.0
Bentonite BB	0.0	0.0	0.0

Bit/BHA Information

BHA No.	Size (in)	Make	Model	Serial Number	Bit Type	IADC Codes	Nozzles (/32")
11	8 1/2	Varel	VM713GUX	6006868	PDC	M332	14/14/14/14
TFA (in²)	**Drilling Time (hrs)**	**Bit Revs**	**Depth In (mKB)**	**Depth Out (mKB)**	**Depth Drilled (m)**	**BHA ROP (m/hr)**	**IADC Bit Dull**
0.60	23.45	299,566.80	2,949.00	3,232.00	283.00	12.1	1-1-WT-G/S-X-1/8-NO-TD
Flow Rate (min) (gpm)	**Flow Rate (max) (gpm)**	**SPP (min) (psi)**	**SPP (max) (psi)**	**WOB (min) (1000lbf)**	**WOB (max) (1000lbf)**	**RPM (min) (rpm)**	**RPM (max) (rpm)**
500	550	2,150.0	2,600.0	5	25	170	214
Circulating Time (hrs)	**Rotating Time (hrs)**	**% Rotating Time (%)**	**% Sliding Time (%)**	**PU HL (max) (1000lbf)**	**Rot HL (max) (1000lbf)**	**SO HL (min) (1000lbf)**	**String Wt (1000lbf)**
33.90	23.45	100.00	0.00	470	378	301	

Drilling Parameters

Depth Start (mKB)	Depth End (mKB)	Drilling Time (hrs)	Drilling Torque	Flow Rate (gpm)	RPM (rpm)	SPP (psi)	WOB (1000lbf)	Drill Str Wt (1000lbf)

Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
3 jts 5 7/8" HWPD	5 7/8	4.000	27.89	252.82
7" Drilling Jar (COR-3021AD)	7	2.938	5.80	224.93
20 jts 5 7/8" HWDP	5 7/8	4.000	186.35	219.13
X/O	6 1/2	2.813	0.85	32.78
6 3/4" Float sub	6 3/4		0.81	31.93
6 3/4" NMDC	6 3/4	3.250	9.48	31.12
6 3/4" TM - Sub	6 3/4		1.74	21.64

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Drill String Components

Item Description	OD (in)	ID (in)	Len (m)	Cum Len (m)
6 3/4" NMDC	6 3/4	3.250	9.48	19.90
8" St. Stabilizer (COR-2941AD)	8	3.000	2.14	10.42
6 3/4" Motor (AKO: 1.25° , Sleeve: 8 3/8")	6 3/4		8.00	8.28
8 1/2" PDC bit	8 1/2		0.28	0.28

Well Control

No.	Depth (mKB)	Stroke (in)	Strokes (spm)	Pres (psi)	Eff (%)	Q (gpm)

Test Date	Last Casing String Run	Test Type	Depth (mKB)	TVD (mKB)	Applied Surface Pressure (psi)
4/8/2014	18 5/8" Shoe, 870.00mKB	Formation Integrity	870.00	843.25	400.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
400.0	8.0	400.00		11.48	1,650.4
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
8/21/2014	13 3/8" Shoe, 1,487.00mKB	Formation Integrity	1,495.00	1,427.35	1,200.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,200.0	7.7	1,200.00		14.03	3,413.7
Test Date	**Last Casing String Run**	**Test Type**	**Depth (mKB)**	**TVD (mKB)**	**Applied Surface Pressure (psi)**
9/5/2014	9 5/8" Shoe, 2,720.80mKB	Formation Integrity	2,728.00	2,583.00	1,540.0
Applied Surface Pressure (psi)	**Volume Pumped (bbl)**	**Current MAASP**	**Kick Tolerance**	**Dens Fluid (lb/gal)**	**Leak Off Pressure (psi)**
1,540.0	5.7	1,540.00		14.00	6,162.4

Geology Information

Formation Name	Prog Top MD (mKB)	Final Top MD (mKB)	Final Top TVD (mKB)
K1		2,732.00	2,586.77
K2		2,851.80	2,696.55
K3		2,901.00	2,740.86
K4		3,029.50	2,856.03
Nar		3,198.50	3,007.23

General Remarks

Comment

>> Requirements: (200 sxs Starch f/ MR #14)

>>> Sent:

** Borkan to ABAN 8: (HPOGC 1x Transmitter + 1 x gamma probe)

>>> Received:

** Nil